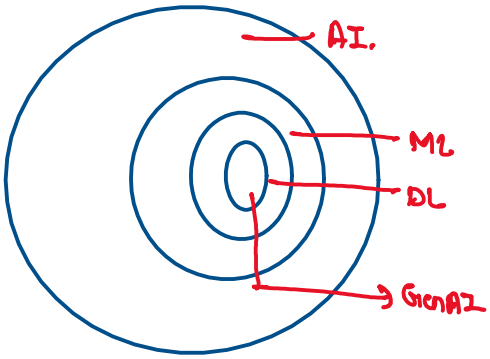
Introduction To Gen AI – 23rd March

Monday, 23 March 2026 6:08 PM

Phase-1 → Python.
FastAPI.

Phase-2 → NLP.
Transformers.
GenAI
- LLMs.
- Work LLMs API
- Key parameters for content Generation
- Prompt engineering.
- Practical Implementation.

- What is Generative AI ?
  ↳ class of models
  - Capable of generating novel & creative content.
  - Produce output such as.
    - Text
    - Images.
    - Codes
    - Audio.



- How GenAI is different from Traditional ML / DL?
  ↳ focus. and capabilities.

Core Distinction

- Traditional ML/DL focused more on prediction, classification or regression task.
- GenAI models focuses on creating new / original content.

- Role of transformers.
  ↳ GPT
  ↳ BERT.

DL → Transfer Learning. → pre-trained model → fine tuning. → desired output
↓ trained on Google
↳ train the model

trained on public data. → billions.

trained on specific data. → small (2 lakh - 5 lakh to lakh).

DL → neural network to capture relationship b/w data.

Based on the data.

Tabular. (row / column) → ANN → Classification / Regression

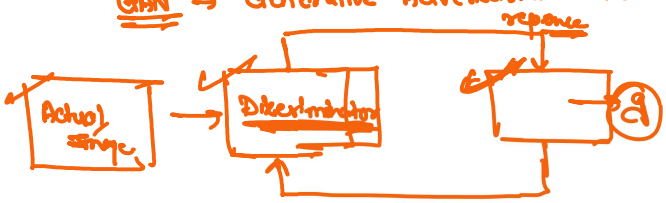
Image. → pixel. — → CNN → Image detection, Image classification, Object Identification

Textual → RNN → Generate sequence
- LSTM → generation
- GRU. → generation
- Encoder-Decoder → generation

→ generate Sequential processing.

→ parallel processing.

GAN → Generative Adversarial network



LLMs. → Generative AI

→ Based on Learned patterns & Input prompt

- Components Involved.
  - Input (Prompt / Instructions).
  - Pre-trained model (LLMs).
  - Output

Solve

↳ halluninate → Complex task

$\int \sin = \cos.$

Example

Solve – halluninate 90%.

$$\int \cos x \, . = -\sin x \, .$$